Answer all questions.

Q1. [50 points]

Data based RL controller implementing PPO algorithm is implemented in rl-wind.ipynb (See RL/LAB). Get familiar with code and see how the different parts – state, observation, rewards, and algorithm are integrated.

Consider the given algorithm as base line. Modify the reward function or power generated equation or both. Display all the parameters as shown in the .ipynb and compare your results with the baseline. The changes in reward function or power equation should be based on physics and consistent with physics/mathematics.

While presentation explain the states, observation, reward function and relevant parameters in detail

Q2. Choose ONLY ONE OPTION [50 points]

Option1 :

Leverage cyberboticss webots (as in Assignment 2) to learn a controller that learns to navigate autonomously with collision avoidance and reaches the predefined goal. Train the model and save it. Then leverage this model to test 5 times – if agent can reach the goal each time. Plot and compare the trajectories as the agent travels for each test.

While presentation explain the environment, states, observation, reward function and relevant parameters in detail. Also explain the interface – how webots is connected to stable baseline - PPO algorithm.

Or

Option 2:

Choose the simulation environment of your choice – Gym, Pybullet or others as explained in class, choose your own robot/agent e.g. Turtlebot, Clear path Husky or Lite 3 or GO to navigate autonomously with collision avoidance. Train the model and save it. Then leverage this model to test 5 times – if agent can reach the goal each time. Plot and compare the trajectories as the agent travels for each test.

While presentation explain the environment, states, observation, reward function and relevant parameters in detail. Also explain the interface – how webots is connected to stable baseline - PPO algorithm.